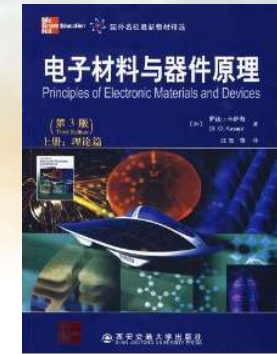
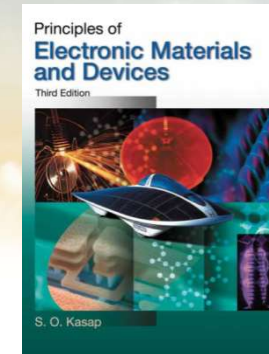




南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》 Solid-State Electronics



Ch2-3 Electrical & Thermal Conduction in Solids

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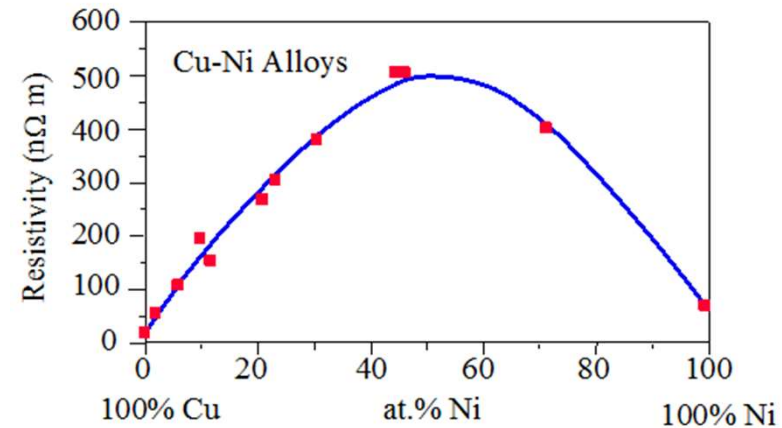
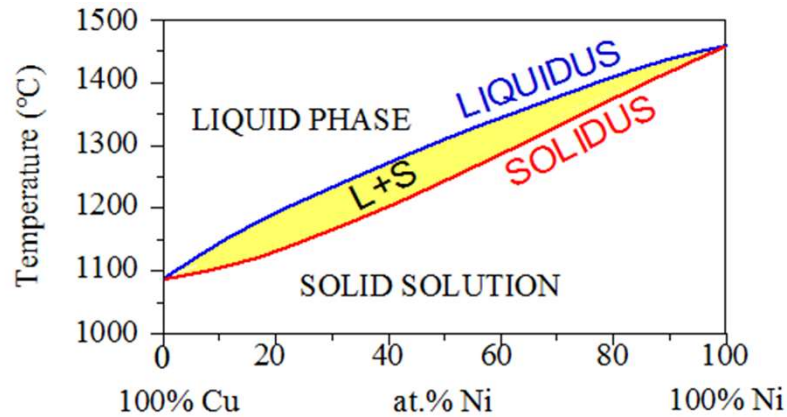
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Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

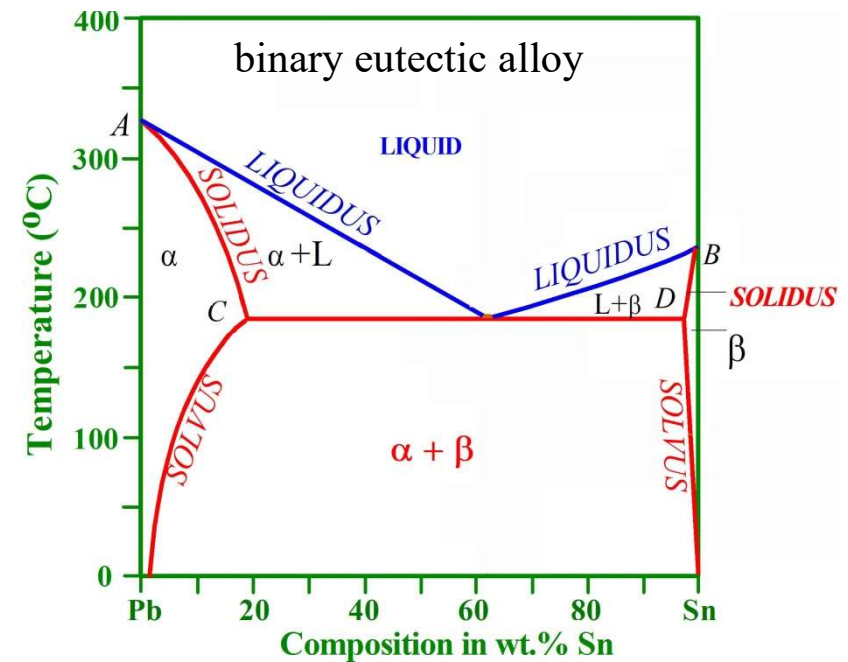


$$\rho = \rho_{matrix} + CX(1 - X)$$

$$= \rho_T + \rho_R + CX(1 - X)$$

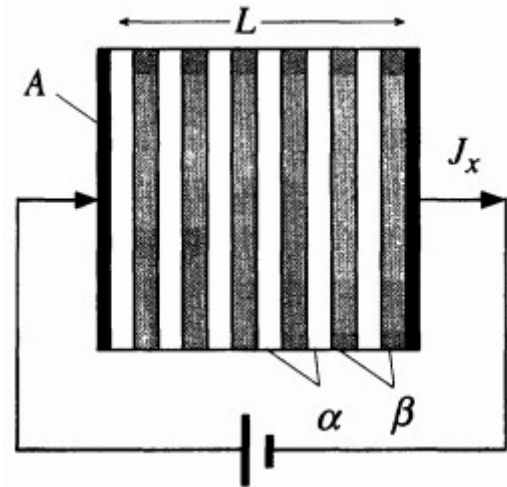
Matthiessen Nordheim

Only applies to solid solutions
that are single-phase solids



2.4 Resistivity of Mixtures and Porous Materials

2.4.1 Heterogeneous Mixtures (多相混合物)



$$R_{\text{eff}} = \frac{L_{\alpha} \rho_{\alpha}}{A} + \frac{L_{\beta} \rho_{\beta}}{A}$$

$$= \frac{L \rho_{\text{eff}}}{A}$$

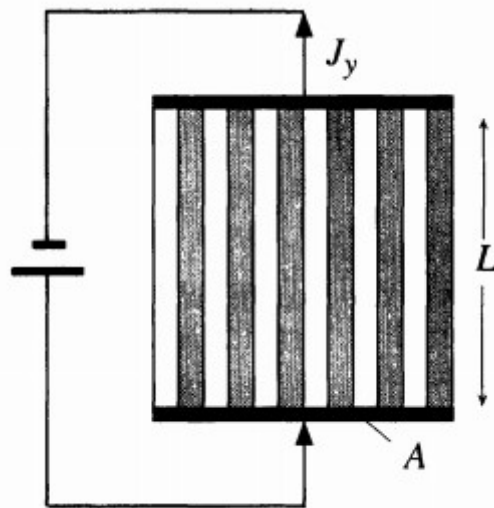
$$L_{\alpha} + L_{\beta} = L$$

$$\chi_{\alpha} = L_{\alpha} / L \quad \chi_{\beta} = L_{\beta} / L$$

Volume fraction
体积分数

$$\rho_{\text{eff}} = \chi_{\alpha} \rho_{\alpha} + \chi_{\beta} \rho_{\beta}$$

Resistivity mixture rule (series rule of mixtures)
电阻率混合物法则 (混合物串联法则)



$$\sigma_{\text{eff}} = \chi_{\alpha} \sigma_{\alpha} + \chi_{\beta} \sigma_{\beta}$$

Conductivity mixture rule (parallel rule of mixtures)
电导率混合物法则 (混合物并联法则)



EXAMPLE 2.13

THE RESISTIVITY-MIXTURE RULE Consider a two-phase alloy consisting of phase α and phase β randomly mixed as shown in Figure 2.14a. The solid consists of a random mixture of two types of resistivities, ρ_α of α and ρ_β of β . We can divide the solid into a bundle of N parallel fibers of length L and cross-sectional area A/N , as shown in Figure 2.14b. In this fiber (infinitesimally thin), the α and β phases are in series, so if $\chi_\alpha = V_\alpha / V$ is the volume fraction of phase α and χ_β is that of β , then the total length of all α regions present in the fiber is $\chi_\alpha L$, and the total length of β regions is $\chi_\beta L$. The two resistances are in series, so the fiber resistance is

$$R_{\text{fiber}} = \frac{\rho_\alpha (\chi_\alpha L)}{(A/N)} + \frac{\rho_\beta (\chi_\beta L)}{(A/N)}$$

But the resistance of the solid is made up of N such fibers in parallel, that is,

$$R_{\text{solid}} = \frac{R_{\text{fiber}}}{N} = \frac{\rho_\alpha \chi_\alpha L}{A} + \frac{\rho_\beta \chi_\beta L}{A}$$

By definition, $R_{\text{solid}} = \rho_{\text{eff}} L / A$, where ρ_{eff} is the effective resistivity of the material, so

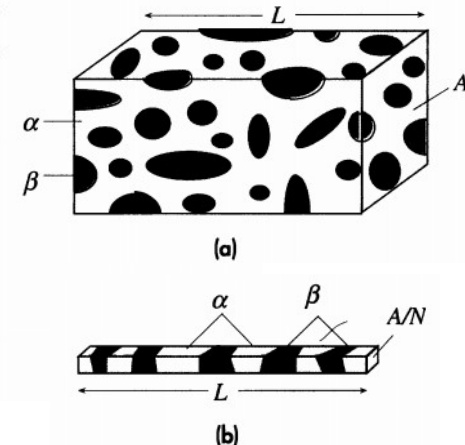
$$\frac{\rho_{\text{eff}} L}{A} = \frac{\rho_\alpha \chi_\alpha L}{A} + \frac{\rho_\beta \chi_\beta L}{A}$$

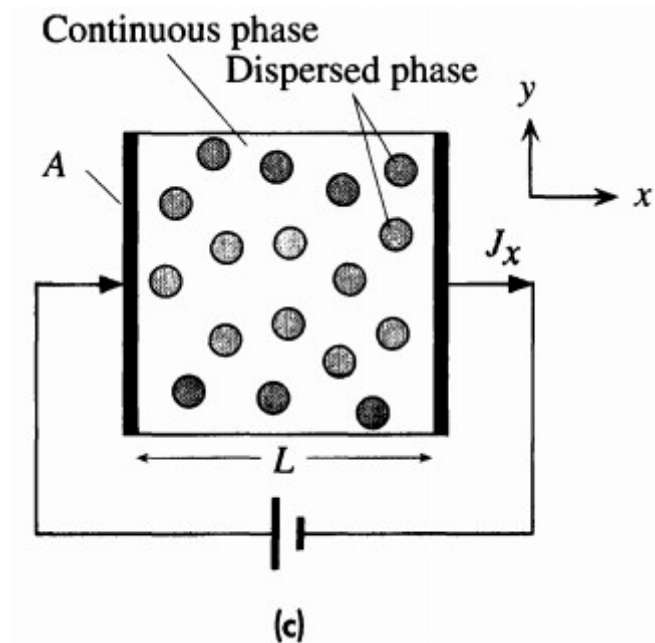
Thus, for a two-phase solid, the effective resistivity will be

$$\rho_{\text{eff}} = \chi_\alpha \rho_\alpha + \chi_\beta \rho_\beta$$

*Resistivity
mixture rule*

If the densities of the two phases are not too different, we can use weight fractions instead of volume fractions. The series rule fails when the resistivities of the phases are vastly different. A major (and critical) tacit assumption here is that the current flow lines are all parallel, so that no current crosses from one fiber to another. Only then can we say that the effective resistance is R_{fiber} / N .





Mixture rule (semi-empirical rules)
混合法则 (半经验公式)

$$\rho_{\text{eff}} = \rho_c \frac{\left(1 + \frac{1}{2} \chi_d\right)}{(1 - \chi_d)} \quad (\rho_d > 10 \rho_c) \quad 2.26$$

$$\rho_{\text{eff}} = \rho_c \frac{(1 - \chi_d)}{(1 + 2 \chi_d)} \quad (\rho_d < 0.1 \rho_c) \quad 2.27$$

EXAMPLE 2.14

A COMPONENT WITH DISPERSED AIR PORES What is the effective resistivity of 95/5 (95% Cu–5% Sn) bronze, which is made from powdered metal containing dispersed pores at 15% (volume percent, vol.%). The resistivity of 95/5 bronze is $1 \times 10^{-7} \Omega \text{ m}$.

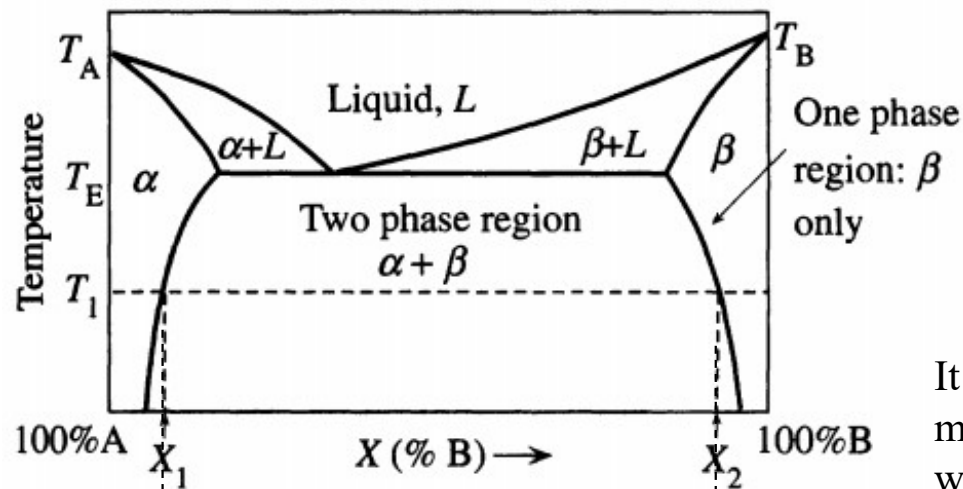
SOLUTION

Pores are infinitely more resistive ($\rho_d = \infty$) than the bronze matrix, so we use Equation 2.26,

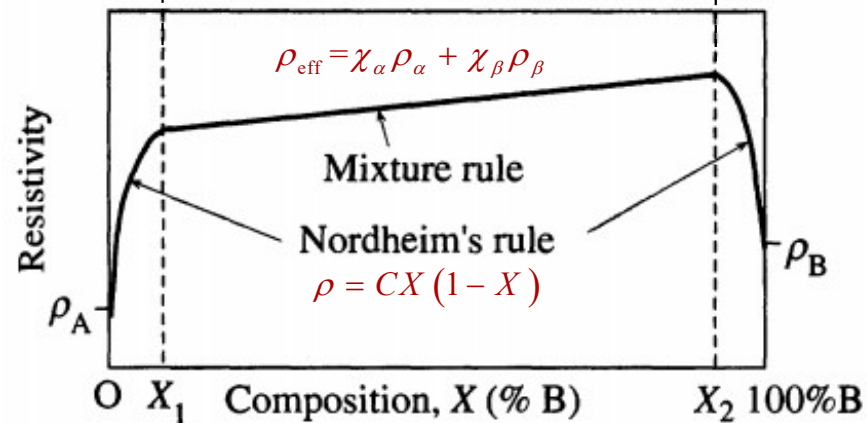
$$\rho_{\text{eff}} = \rho_c \frac{1 + \frac{1}{2} \chi_d}{1 - \chi_d} = (1 \times 10^{-7} \Omega \text{ m}) \frac{1 + \frac{1}{2}(0.15)}{1 - 0.15} = 1.27 \times 10^{-7} \Omega \text{ m}$$



2.4.2 Two-Phase Alloy Resistivity and Electrical Constant



(a)



(b)

Figure 2.15 Eutectic-forming alloys, e.g., Cu–Ag.

(a) The phase diagram for a binary, eutectic-forming alloy.

(b) The resistivity versus composition for the binary alloy.

It is frequently necessary to improve the mechanical properties of a metal alloy without significantly impairing its electrical conductivity.

Solid-solution alloying improves mechanical strength, but at the expense of conductivity.

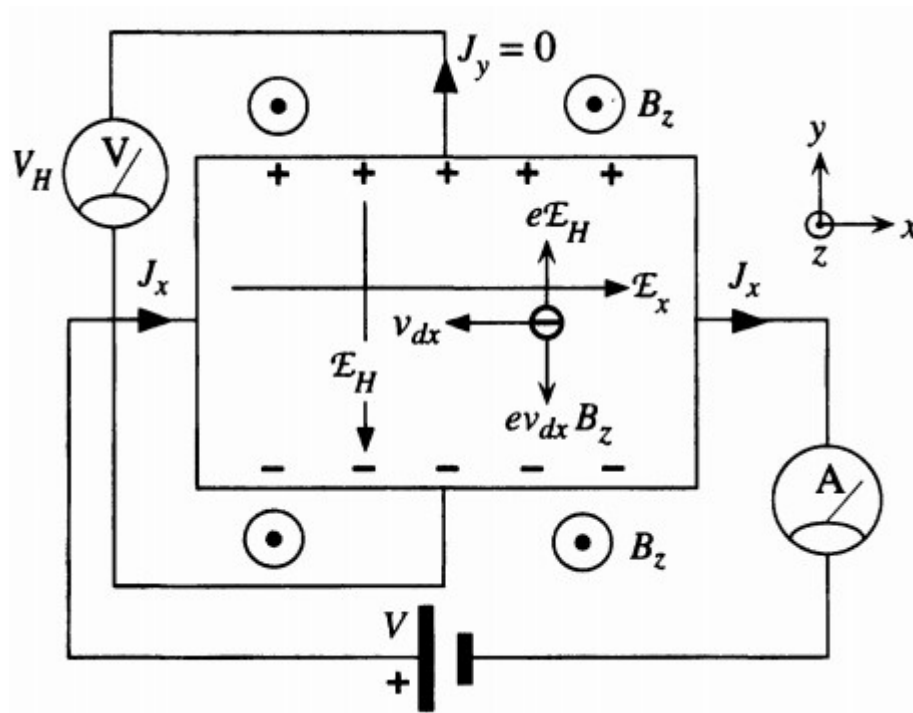
A compromise must often be found between electrical and mechanical properties.

Most often, strength is enhanced by introducing a second phase that does not have such an adverse effect on the conductivity.



2.5 The Hall Effect and Hall Devices

The Hall Effect



$$\vec{F} = q\vec{v} \times \vec{B}$$

$$\left. \begin{aligned} eE_H &= ev_{dx}B_z \\ J_x &= env_{dx} \end{aligned} \right\} E_H = \left(\frac{1}{en} \right) J_x B_z$$

Definition of Hall coefficient

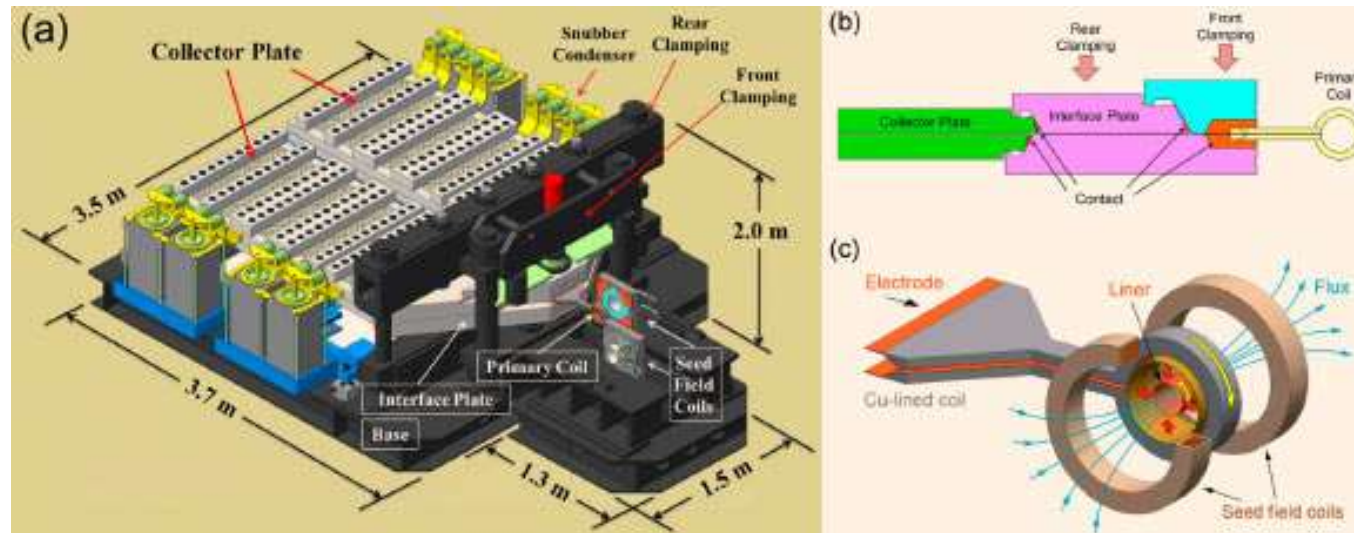
$$R_H = \frac{E_y}{J_x B_z} = -\frac{1}{en}$$

Hall coefficient for electron conduction

Q: Which one possesses more significant Hall effect? Metal or **Semiconductor**?



Japanese Physicists Generate Strongest Magnetic Field Ever Achieved Indoors



Physicists from the Institute for Solid State Physics at the University of Tokyo, Japan, have recorded the largest magnetic field ever generated indoors — a whopping 1,200 T (tesla).

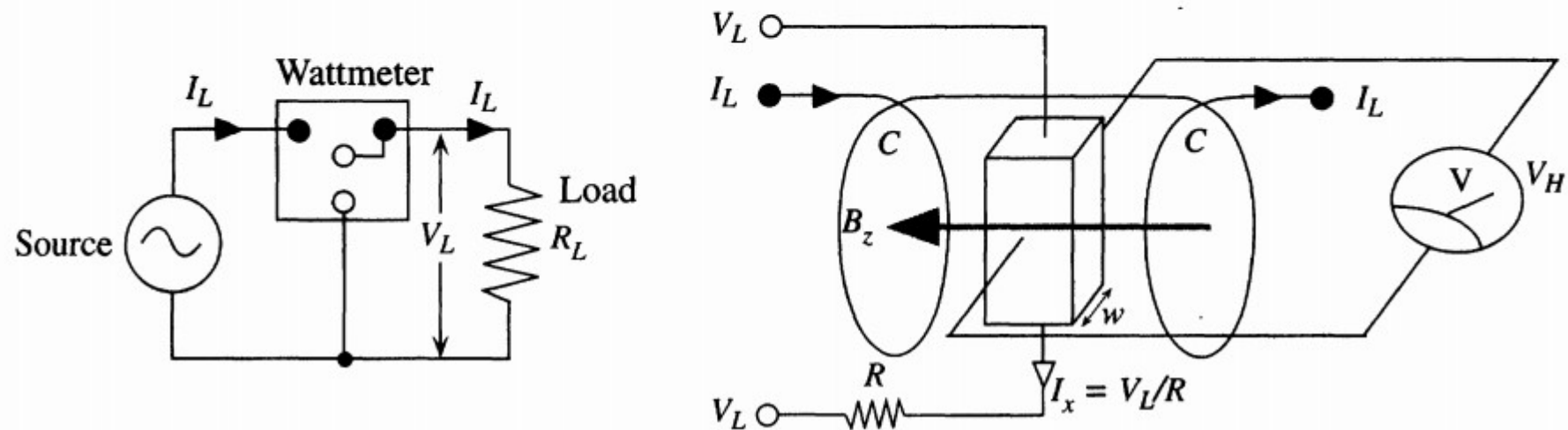
Sep-17-2018



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EXAMPLE 2.16

HALL-EFFECT WATTMETER The Hall effect can be used to implement a wattmeter to measure electrical power dissipated in a load. The schematic sketch of the Hall-effect wattmeter is shown in Figure 2.18, where the Hall-effect sample is typically a semiconductor material (usually Si). The load current I_L passes through two coils, which are called current coils and are shown as C in Figure 2.18. These coils set up a magnetic field B_z such that $B_z \propto I_L$. The Hall-effect sample is positioned in this field between the coils. The voltage V_L across the load drives a current



$$V_H = wE_H = wR_H J_x B_z \propto I_x B_z \propto V_L I_L$$



EXAMPLE 2.17

HALL MOBILITY Show that if R_H is the Hall coefficient and σ is the conductivity of a metal, then the drift mobility of the conduction electrons is given by

$$\mu_d = |\sigma R_H| \quad [2.33]$$

The Hall coefficient and conductivity of copper at 300 K have been measured to be $-0.55 \times 10^{-10} \text{ m}^3 \text{ A}^{-1} \text{ s}^{-1}$ and $5.9 \times 10^7 \text{ } \Omega^{-1} \text{ m}^{-1}$, respectively. Calculate the drift mobility of electrons in copper.

SOLUTION

Consider the expression for

$$R_H = \frac{-1}{en}$$

Since the conductivity is given by $\sigma = en\mu_d$, we can substitute for en to obtain

$$R_H = \frac{-\mu_d}{\sigma} \quad \text{or} \quad \mu_d = -R_H \sigma$$

which is Equation 2.33. The drift mobility can thus be determined from R_H and σ .

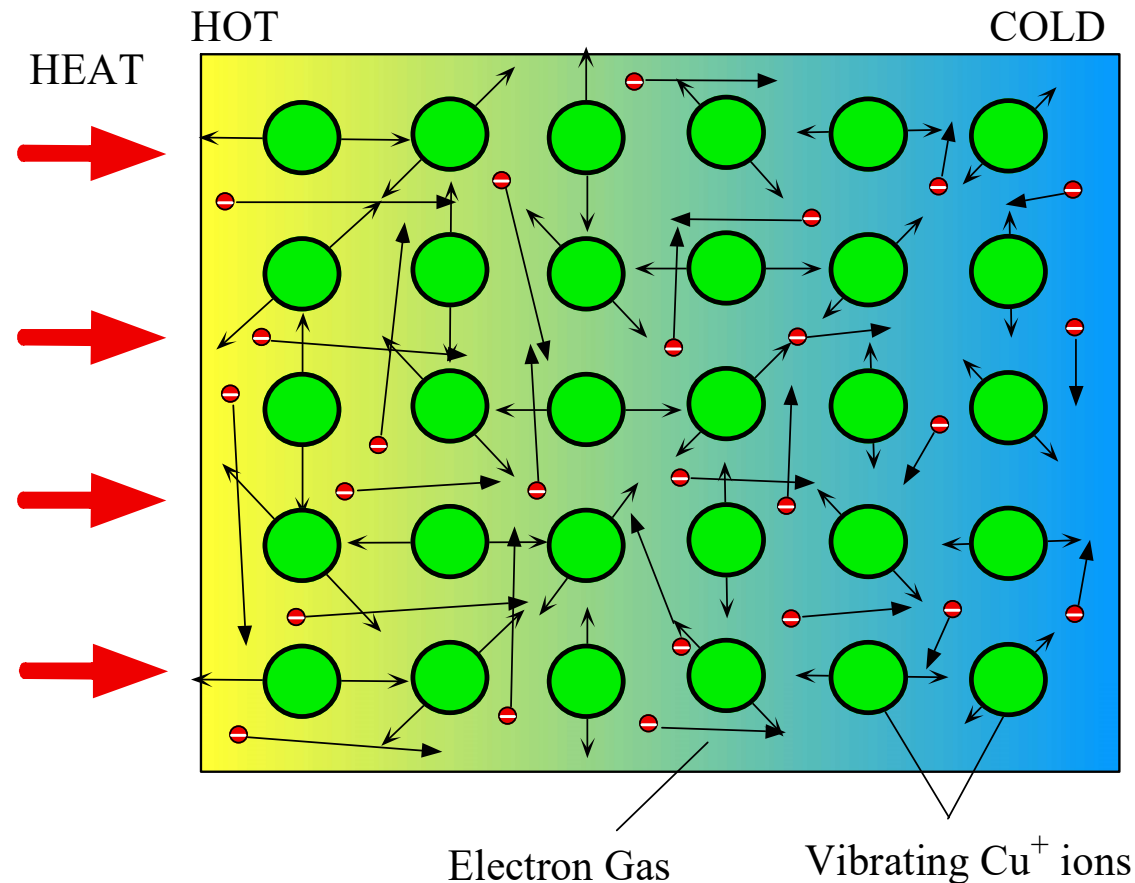
The product of σ and R_H is called the **Hall mobility** μ_H . Some values for the Hall mobility of electrons in various metals are listed in Table 2.4. From the expression in Equation 2.33, we get

$$\mu_d = -(-0.55 \times 10^{-10} \text{ m}^3 \text{ A}^{-1} \text{ s}^{-1})(5.9 \times 10^7 \text{ } \Omega^{-1} \text{ m}^{-1}) = 3.2 \times 10^{-3} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

It should be mentioned that Equation 2.33 is an oversimplification. The actual relationship involves a numerical factor that multiplies the right term in Equation 2.33. The factor depends on the charge carrier scattering mechanism that controls the drift mobility.

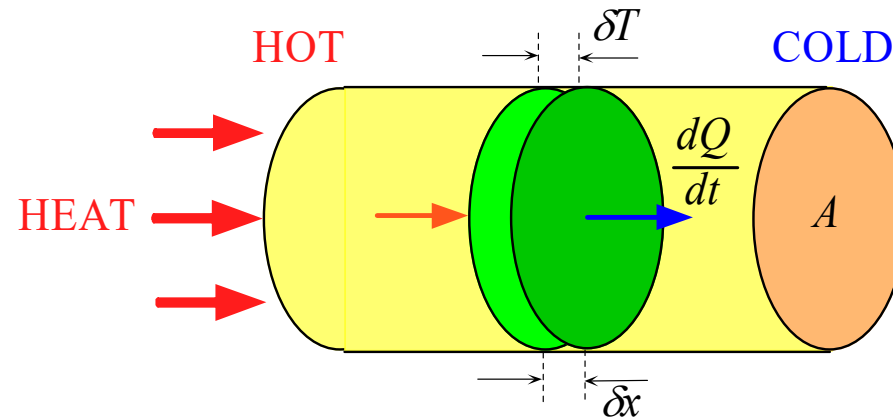


2.6 Thermal Conduction (热导)



Thermal conduction in a metal involves transferring energy from the hot region to the cold region by conduction electrons. More energetic electrons (shown with longer velocity vectors) from the hotter regions arrive at cooler regions and collide there with lattice vibrations and transfer their energy. Lengths of arrowed lines on atoms represent the magnitudes of atomic vibrations.





Heat flow in a metal rod heated at one end. Consider the rate of heat flow, dQ/dt , across a thin section δx of the rod. The rate of heat flow is proportional to the temperature gradient $\delta T/\delta x$ and the cross sectional area A .

The rate of heat flow $Q' = dQ/dt$

$$Q' = -A\kappa \frac{\delta T}{\delta x}$$

$$I = -A\sigma \frac{\delta V}{\delta x}$$

κ Constant of proportionality;
thermal conductivity
热导率

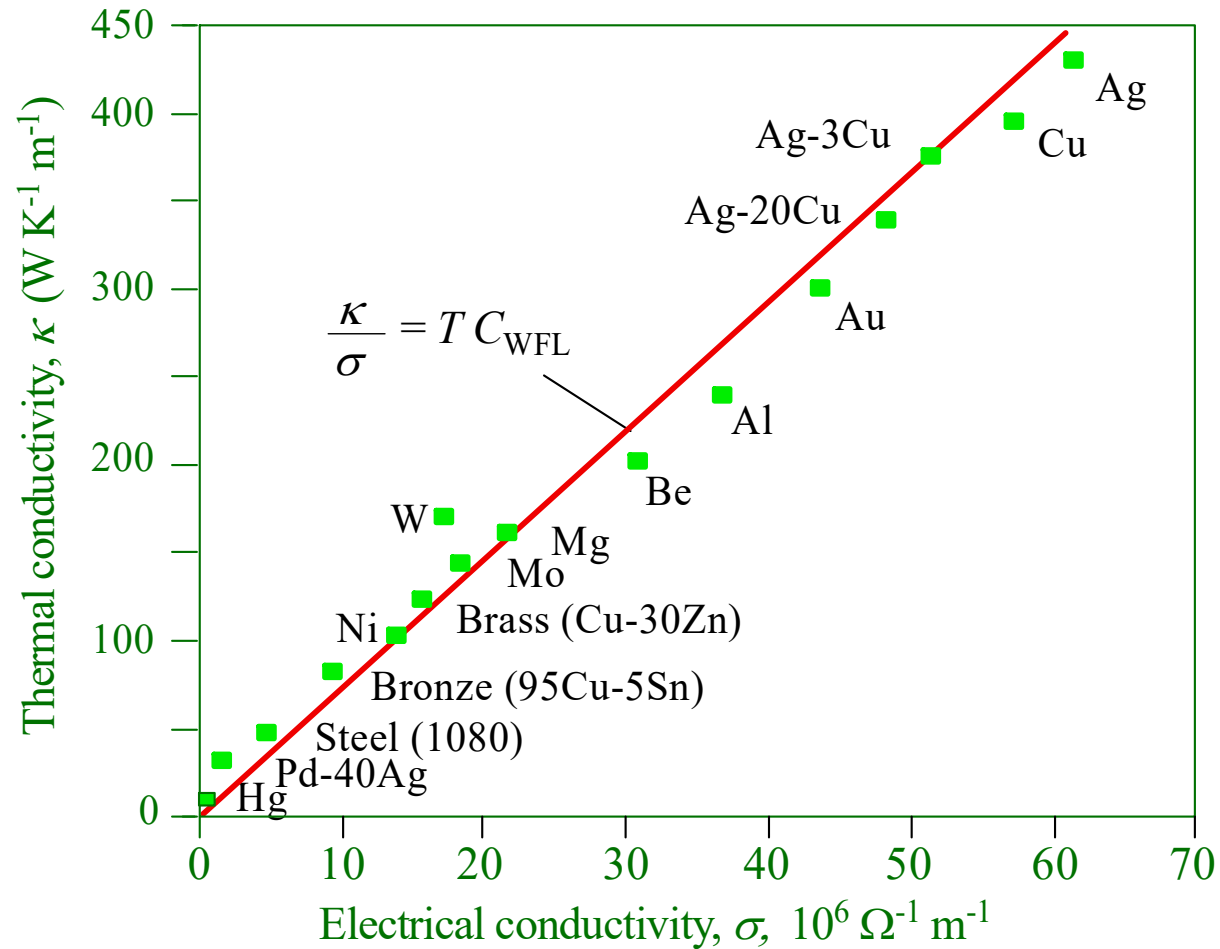
$\frac{\delta T}{\delta x}$ Temperature gradient
温度梯度

$$\frac{\kappa}{\sigma T} = C_{\text{WFL}}$$

$$C_{\text{WFL}} = \frac{\pi^2 k^2}{3e^2} = 2.44 \times 10^{-8} \text{ W} \cdot \Omega \cdot \text{K}^{-2}$$

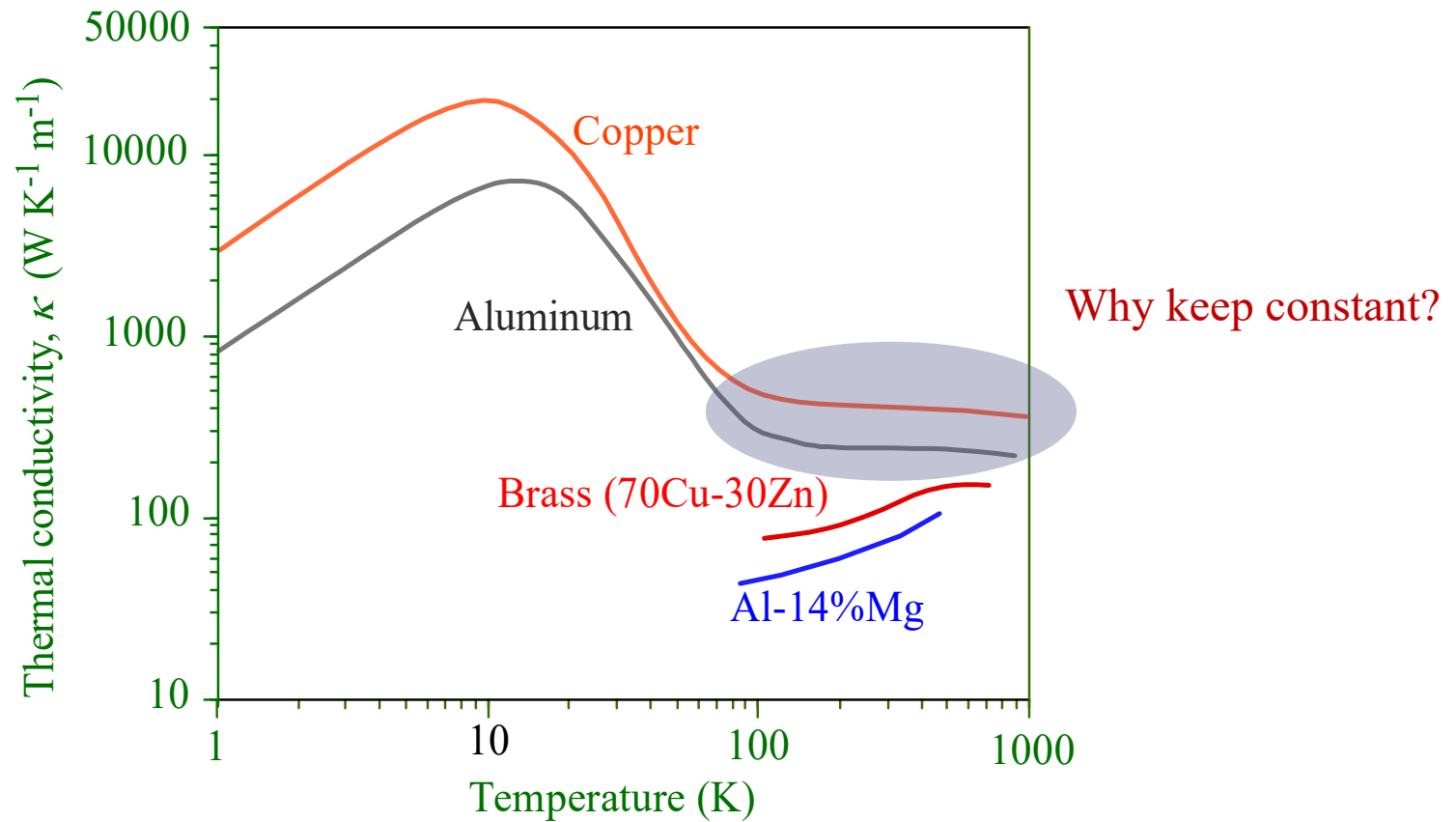
Lorenz number (洛伦兹数)
Wiedemann-Franz-Lorenz Law





Thermal conductivity, κ vs. electrical conductivity σ for various metals (elements and alloys) at 20 °C. The solid line represents the WFL law with $C_{\text{WFL}} = 2.44 \times 10^{-8} \text{ W} \cdot \Omega \cdot \text{K}^{-2}$.

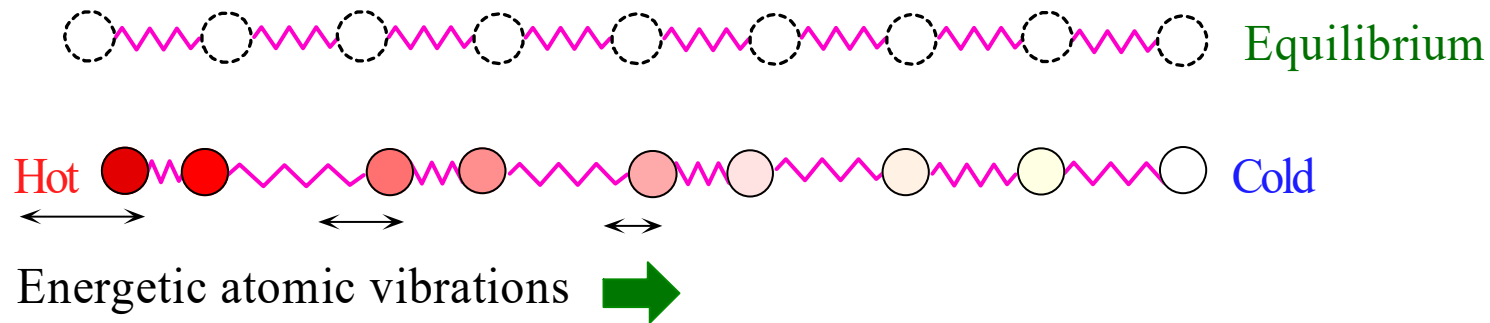




Thermal conductivity vs. temperature for two pure metals (Cu and Al) and two alloys (brass and Al-14%Mg). Data extracted from *Thermophysical Properties of Matter*, Vol. 1: Thermal Conductivity, Metallic Elements and Alloys, Y.S. Touloukian et. al (Plenum, New York, 1970).



Thermal Conductivity of Non-Metallic Materials (非金属的热导)



Conduction of heat in insulators involves the generation and propagation of atomic vibrations through the bonds that couple the atoms. (An intuitive figure.)

The effectiveness of heat transfer and the coupling efficiency between atoms, the propagation of vibrational waves in crystals, and the scattering of crystal defects have been related to their interaction with other vibrational waves.

Q: If the material has **strong** or weak covalent bonds, which material has higher thermal conductivity?



Table 2.5 Typical thermal conductivities of various classes of materials at 25 °C

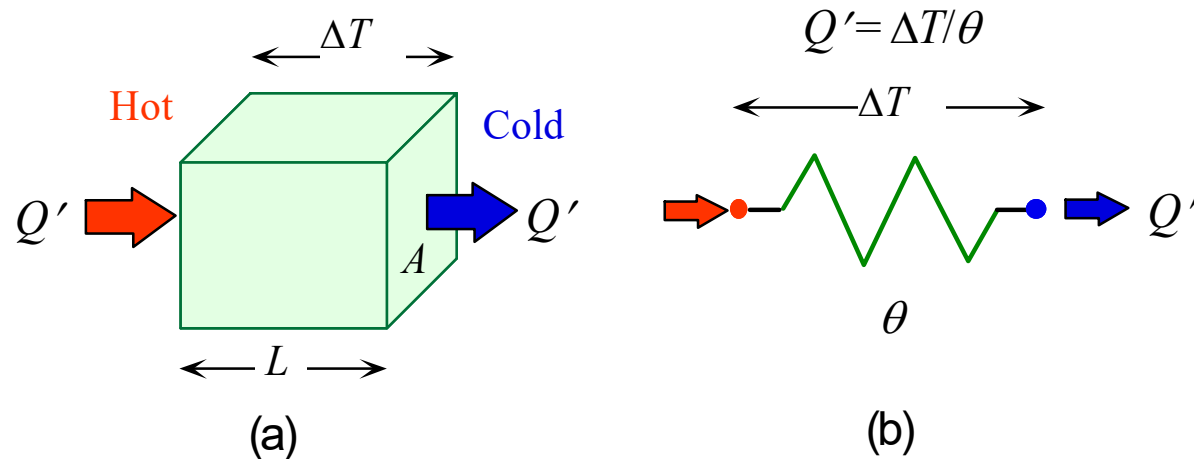
Material	κ (W m ⁻¹ K ⁻¹)
Pure metal	
Nb	52
Fe	80
Zn	113
W	178
Al	250
Cu	390
Ag	420
Metal alloys	
Stainless steel	12–16
55% Cu–45% Ni	19.5
70% Ni–30% Cu	25
1080 steel	50
Bronze (95% Cu–5% Sn)	80
Brass (63% Cu–37% Zn)	125
Dural (95% Al–4% Cu–1% Mg)	147
Ceramics and glasses	
Glass-borosilicate	0.75
Silica-fused (SiO ₂)	1.5
S ₃ N ₄	20
Alumina (Al ₂ O ₃)	30
Sapphire (Al ₂ O ₃)	37
Beryllium (BeO)	260
Diamond	~1000
Polymers	
Polypropylene	0.12
PVC	0.17
Polycarbonate	0.22
Nylon 6,6	0.24
Teflon	0.25
Polyethylene, low density	0.3
Polyethylene, high density	0.5

续表 2.5

材料	κ (W · m ⁻¹ · K ⁻¹)
青铜(95%铜-5%锡)	80
黄铜(63%铜-37%锌)	125
硬铝(95%铝-4%铜-1%镁)	147
陶瓷与玻璃	
硼硅酸盐玻璃	0.75
熔融石英(SiO ₂)	1.5
氮化硅(S ₃ N ₄)	20
氧化铝(Al ₂ O ₃)	30
蓝宝石(Al ₂ O ₃)	37
氧化铍(BeO)	260
金刚石	~1000
聚合物	
聚丙烯	0.12
聚氯乙烯	0.17
聚碳酸酯	0.22
尼龙 6,6	0.24
聚四氟乙烯	0.25
低密度聚乙烯	0.3
高密度聚乙烯	0.5



2.6.2 Thermal resistance (热阻)



Conduction of heat through a component in (a) can be modeled as a thermal resistance θ shown in (b) where $Q' = \Delta T / \theta$.

$$Q' = A\kappa \frac{\Delta T}{L} = \frac{\Delta T}{\left(\frac{L}{\kappa A}\right)} = \frac{\Delta T}{\theta}$$

$$I = A\sigma \frac{\Delta V}{L} = \frac{\Delta V}{\left(\frac{L}{\sigma A}\right)} = \frac{\Delta V}{R}$$

θ : thermal resistance 热阻

R : resistance 电阻



EXAMPLE 2.20

THERMAL RESISTANCE A brass disk of electrical resistivity $50 \text{ n}\Omega \text{ m}$ conducts heat from a heat source to a heat sink at a rate of 10 W . If its diameter is 20 mm and its thickness is 30 mm , what is the temperature drop across the disk, neglecting the heat losses from the surface?

SOLUTION

We first determine the thermal conductivity:

$$\begin{aligned}\kappa &= \sigma TC_{\text{WFL}} = (5 \times 10^{-8} \Omega \text{ m})^{-1} (300 \text{ K}) (2.44 \times 10^{-8} \text{ W } \Omega \text{ K}^{-2}) \\ &= 146 \text{ W m}^{-1} \text{ K}^{-1}\end{aligned}$$

The thermal resistance is

$$\theta = \frac{L}{\kappa A} = \frac{(30 \times 10^{-3} \text{ m})}{\pi (10 \times 10^{-3} \text{ m})^2 (146 \text{ W m}^{-1} \text{ K}^{-1})} = 0.65 \text{ K W}^{-1}$$

Therefore, the temperature drop is

$$\Delta T = \theta Q' = (0.65 \text{ K W}^{-1}) (10 \text{ W}) = 6.5 \text{ K or } ^\circ\text{C}$$

